

Logistic Regression

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1 Derivation

1.1 Sigmoid Function and $P(y = 1|\mathbf{x}; \theta)$

In Logistic Regression what we want to optimize is the probability that an observation belongs to a certain class (0 or 1), we will focus on optimizing for class 1 given it is a binary classification problem, optimizing for one also optimizes for the other. In order to model a probability between 0 and 1, we will use the sigmoid function which squishes the entire real numbers line $(-\infty, \infty)$ to $(0, 1)$ as shown in figure 1.

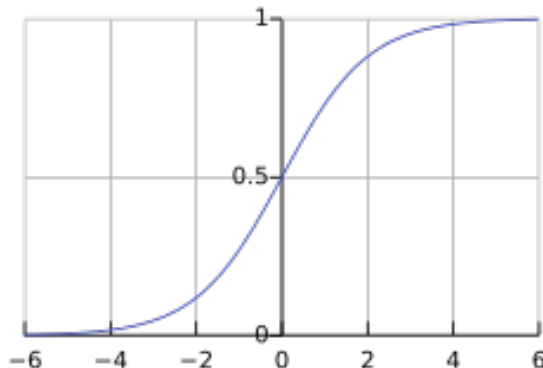


Figure 1: Sigmoid Function

So given a feature vector with one bias term $x_0 = 1$, we have $\mathbf{x}_i \in R^{n+1}$ for the i th training example and a weight vector $\theta \in R^{n+1}$, we can write down the sigmoid function $\sigma(z)$ for this task as:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (1)$$

$$\sigma(\theta^T \mathbf{x}) = \frac{1}{1 + e^{\theta^T \mathbf{x}}} \quad (2)$$

Where $\sigma(\theta^T \mathbf{x}) = P(y = 1|\mathbf{x}; \theta)$ with $P(y = 0|\mathbf{x}; \theta) = 1 - P(y = 1|\mathbf{x}; \theta)$

1.2 Maximum Likelihood Estimation

To find the best set of parameters that best explain both classes given $\mathbf{x}$, we will first calculate the likelihood of the entire dataset $L(\theta)$ then get the log of this function to convert products to sums, that is $l(\theta)$ which we call the log-likelihood. First, we can write the PMF of this task as a bernoulli distribution given there are only two classes so two distinct probabilities.

$$P(y|\mathbf{x}; \theta) = \sigma^y (1 - \sigma)^{1-y} \quad (3)$$

So the likelihood $L(\theta)$ for the entire dataset becomes the product of all the observations, assuming that each observation is independent and identically distributed (**IID**)

$$L(\theta) = \prod_{i=1}^m [\sigma(\theta^T \mathbf{x}_i)]^{y_i} \cdot [1 - \sigma(\theta^T \mathbf{x}_i)]^{1-y_i} \quad (4)$$

If we take the logarithm of equation 4 then it becomes the log-likelihood which is:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^m \sigma_i^{y_i} \cdot (1 - \sigma_i)^{1-y_i} \\ \log L(\theta) &= \sum_{i=1}^m \log[\sigma_i^{y_i} \cdot (1 - \sigma_i)^{1-y_i}] \\ l(\theta) &= \sum_{i=1}^m [\log(\sigma_i^{y_i}) + \log((1 - \sigma_i)^{1-y_i})] \\ &= \sum_{i=1}^m [y_i \log(\sigma_i) + (1 - y_i) \log(1 - \sigma_i)] \end{aligned}$$

Then we can rewrite the log-likelihood as:

$$l(\theta) = \sum_{i=1}^m [y_i \log \sigma(\theta^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\theta^T \mathbf{x}_i))] \quad (5)$$

Finally, we can rewrite the log-likelihood as our cost function by negating it given finding the maximum of $f(x)$ is the same as finding the minimum of $-f(x)$, we do this because cost functions want to minimize, not maximize.

$$J(\theta) = - \sum_{i=1}^m [y_i \log \sigma(\theta^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\theta^T \mathbf{x}_i))] \quad (6)$$

An optional step is taking the average of the cost function (dividing by the total number of observations m) so that what we minimize doesn't explode to a huge number in computation.

$$J(\theta) = - \frac{1}{m} \sum_{i=1}^m [y_i \log \sigma(\theta^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\theta^T \mathbf{x}_i))] \quad (7)$$

We commonly refer to equation 6 as "Binary Cross Entropy Loss".

Expanding the sigmoid function in equation 7 we can rewrite it as:

$$J(\theta) = - \frac{1}{m} \sum_{i=1}^m [y_i \theta^T \mathbf{x}_i - \log(1 + e^{\theta^T \mathbf{x}_i})] \quad (8)$$

1.3 Gradient Descent

Having our cost function we would like to calculate now both the gradient and Hessian (the hessian for newton's method), starting off with the gradient for gradient descent, leveraging one key property of the derivative of the sigmoid function which is

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1 - \sigma(z)) \quad (9)$$

Which we can prove by taking equation 1 and calculating its derivative

$$\begin{aligned} \sigma(z) &= \frac{1}{1 + e^{-z}} \\ \sigma(z) &= (1 + e^{-z})^{-1} \\ \frac{d\sigma(z)}{dz} &= -(1 + e^{-z})^{-2}(-e^{-z}) \\ \frac{d\sigma(z)}{dz} &= (1 + e^{-z})^{-1}(1 + e^{-z})^{-1}(e^{-z}) \\ \frac{d\sigma(z)}{dz} &= \sigma(z) \frac{e^{-z} + 1 - 1}{1 + e^{-z}} \end{aligned}$$

Therefore the derivative of $\sigma(z)$ is:

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1 - \sigma(z))$$

Now we can calculate the gradient of equation 7, we will begin by calculating the derivative of the cost function w.r.t the j th parameter, afterwards we can generalize for its gradient.

$$\frac{\partial J}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial \theta_j} [y_i \log \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))] \quad (10)$$

For the first part $\frac{\partial}{\partial \theta_j} y_i \log \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)$, using 9 we get via chain rule

$$\begin{aligned} y_i \frac{\partial}{\partial \theta_j} \log \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) &= y_i \frac{1}{\sigma(\boldsymbol{\theta}^T \mathbf{x}_i)} \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \mathbf{x}_i \\ y_i \frac{\partial}{\partial \theta_j} \log \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) &= y_i (1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \mathbf{x}_i \end{aligned}$$

And for the second part $\frac{\partial}{\partial \theta_j} (1 - y_i) \log(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))$ we get

$$\begin{aligned} (1 - y_i) \frac{\partial}{\partial \theta_j} \log(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) &= (1 - y_i) \frac{1}{1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)} (-\sigma'(\boldsymbol{\theta}^T \mathbf{x}_i)) \mathbf{x}_i \\ &= (1 - y_i) \frac{-1}{1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)} \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \mathbf{x}_i \\ &= -(1 - y_i) \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) \mathbf{x}_i \end{aligned}$$

So now we can write the full partial derivative as:

$$\begin{aligned} \frac{\partial J}{\partial \theta_j} &= -\frac{1}{m} \sum_{i=1}^m \mathbf{x}_i y_i (1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) - \mathbf{x}_i (1 - y_i) \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) \\ &= -\frac{1}{m} \sum_{i=1}^m \mathbf{x}_i (y_i - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \end{aligned}$$

So $\frac{\partial J}{\partial \theta_j}$ is equal to

$$\frac{\partial J}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m \mathbf{x}_i^{(j)} (y_i - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \quad (11)$$

where $\mathbf{x}_i^{(j)}$ is the j th feature of the i th example.

So now the gradient of the cost function $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ becomes: $\nabla_{\boldsymbol{\theta}} J = \begin{bmatrix} \frac{\partial J}{\partial \theta_0} \\ \frac{\partial J}{\partial \theta_1} \\ \vdots \\ \frac{\partial J}{\partial \theta_n} \end{bmatrix}$

if we denote $\mathbf{X}$ as our design matrix where each row is an observation and each column is a separate parameter θ_j , we can see that that the summation represents a dot product between $\mathbf{X}^T$ and the error of the whole training set (our prediction). We take the transposed of our matrix given each row is a set of observations and what $\mathbf{x}_i^{(j)}$ represents is the feature for the j th component, so the feature across all m examples. Therefore we can rewrite equation 11 to:

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= \frac{-1}{m} \mathbf{X}^T \cdot (\text{error}) \\ &= \frac{-1}{m} \mathbf{X}^T \cdot (\mathbf{y} - \boldsymbol{\sigma}) \\ \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) &= -\frac{1}{m} \mathbf{X}^T \cdot (\mathbf{y} - \boldsymbol{\sigma}) \end{aligned} \quad (12)$$

So the update rule for gradient descent where alpha is the constant learning rate is:

$$\theta^{i+1} := \theta^i + \alpha \frac{1}{m} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\sigma}) \quad (13)$$

Notice that there is a plus sign instead of a minus sign in the update rule, this is because the minus sign we introduced on the cost function cancels out with the minus sign that is introduced in gradient descent.

1.4 Newton's Method

Now for Newton's method, the formal definition is

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \mathbf{H}^{-1}(\boldsymbol{\theta}_k) \nabla J(\boldsymbol{\theta}_k) \quad (14)$$

The only thing we have to calculate now is the Hessian and its inverse. To calculate the hessian, first we would like to recall the definition of the Hessian.

$$\mathbf{H}f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \quad (15)$$

We have already calculated $\frac{\partial J}{\partial \theta_j}$ in equation 11, so now we only need to differentiate again but now with respect to θ_k to also get the off diagonal elements of the Hessian.

$$\begin{aligned} \frac{\partial^2 J}{\partial \theta_j \partial \theta_k} &= \frac{-1}{m} \sum_{i=1}^m \frac{\partial}{\partial \theta_k} [\mathbf{x}_i (y_i - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))] \\ \mathbf{H}_{jk} &= \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i^{(j)} \mathbf{x}_i^{(k)} \sigma(\boldsymbol{\theta}^T \mathbf{x}_i) (1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)) \end{aligned}$$

Now we want to express the full Hessian $\mathbf{H}$ simultaneously using only vectors and matrices. For the first part which is $\sum_{i=1}^m \mathbf{x}_i^{(j)} \mathbf{x}_i^{(k)}$, we can represent this using our design matrix $\mathbf{X}$ as:

$$\sum_{i=1}^m \mathbf{x}_i^{(j)} \mathbf{x}_i^{(k)} = \mathbf{X}^T \mathbf{X}$$

Where each (j, k) pair entry represents that sum. Now for the last part which is $\sum_{i=1}^m \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))$ this part depends only on the observation we are looking at, which means that for a fixed sample i every entry of its contribution to the Hessian gets the same scalar value. In matrix notation, left-multiplying by a diagonal matrix scales each row which is exactly what we want. If we define our scalar component $\sigma(\boldsymbol{\theta}^T \mathbf{x}_i)(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))$ to be a diagonal matrix $\mathbf{S} = \text{diag}(s_1, \dots, s_m)$ where each entry s_{ii} is each observation's scale factor, we can rewrite the Hessian as:

$$\mathbf{H} = \frac{1}{m} \mathbf{X}^T \mathbf{S} \mathbf{X} \quad (16)$$

where

$$\mathbf{S}_{ii} = \sigma(\boldsymbol{\theta}^T \mathbf{x}_i)(1 - \sigma(\boldsymbol{\theta}^T \mathbf{x}_i))$$

$$\mathbf{S} = \begin{pmatrix} \sigma_1(1 - \sigma_1) & 0 & \cdots & 0 \\ 0 & \sigma_2(1 - \sigma_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_m(1 - \sigma_m) \end{pmatrix} \quad (17)$$

And finally, the update rule for newton's method from equation 14 becomes

$$\theta^{i+1} := \theta^i - \left(\frac{1}{m} \mathbf{X}^T \mathbf{S} \mathbf{X} \right)^{-1} \left(-\frac{1}{m} \mathbf{X}^T \cdot (\mathbf{y} - \boldsymbol{\sigma}) \right)$$

$$\theta^{i+1} := \theta^i + (\mathbf{X}^T \mathbf{S} \mathbf{X})^{-1} \mathbf{X}^T \cdot (\mathbf{y} - \boldsymbol{\sigma}) \quad (18)$$